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Investigating river–aquifer interactions using heat as a tracer in the Silala river transboundary basin

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Abstract

This article reviews scientific studies in which heat was used as a natural tracer to investigate stream–aquifer interactions in the Silala River in Chile and provide evidence that was used to support a legal dispute between Chile and Bolivia over the status and use of the waters of this watercourse. Streambed temperature time series at various locations downstream of the Chile–Bolivia international border showed that water flows downwards through the streambed sediments, from the river towards the fluvial aquifer. These findings are consistent with hydraulic head measurements performed at the study site. Additionally, fiber-optic distributed temperature sensing (FO-DTS) methods were employed to observe river temperatures with a spatial resolution of the order of 0.5 m in a river reach of ~1.3 km. FO-DTS technology allowed detection of various warm springs that discharged their waters into the Silala River, as well as the location of an artesian well that supplies the river with ~90 L/s of deep groundwater at ~20 °C. The results provided improved understanding of the Silala River hydrogeology, and were used to calibrate and validate a groundwater model of the system, reported elsewhere. Both methodologies demonstrated that the river is indeed a system of surface waters and groundwaters interacting as a unitary whole, a key aspect of the definition of an international watercourse, and generated valuable scientific evidence to support a major international legal dispute.

This article is categorized under:

Science of Water > Hydrological Processes

Science of Water > Methods

KEYWORDS

distributed temperature sensing, groundwater–surface waters interaction, thermal methods

1 | INTRODUCTION

The Silala river originates at approximately 4400 m ASL in Bolivian territory where water emerges from groundwater-fed springs in the Orientales and Cajones Ravines, located <4 km northeast of the Chile–Bolivia international border

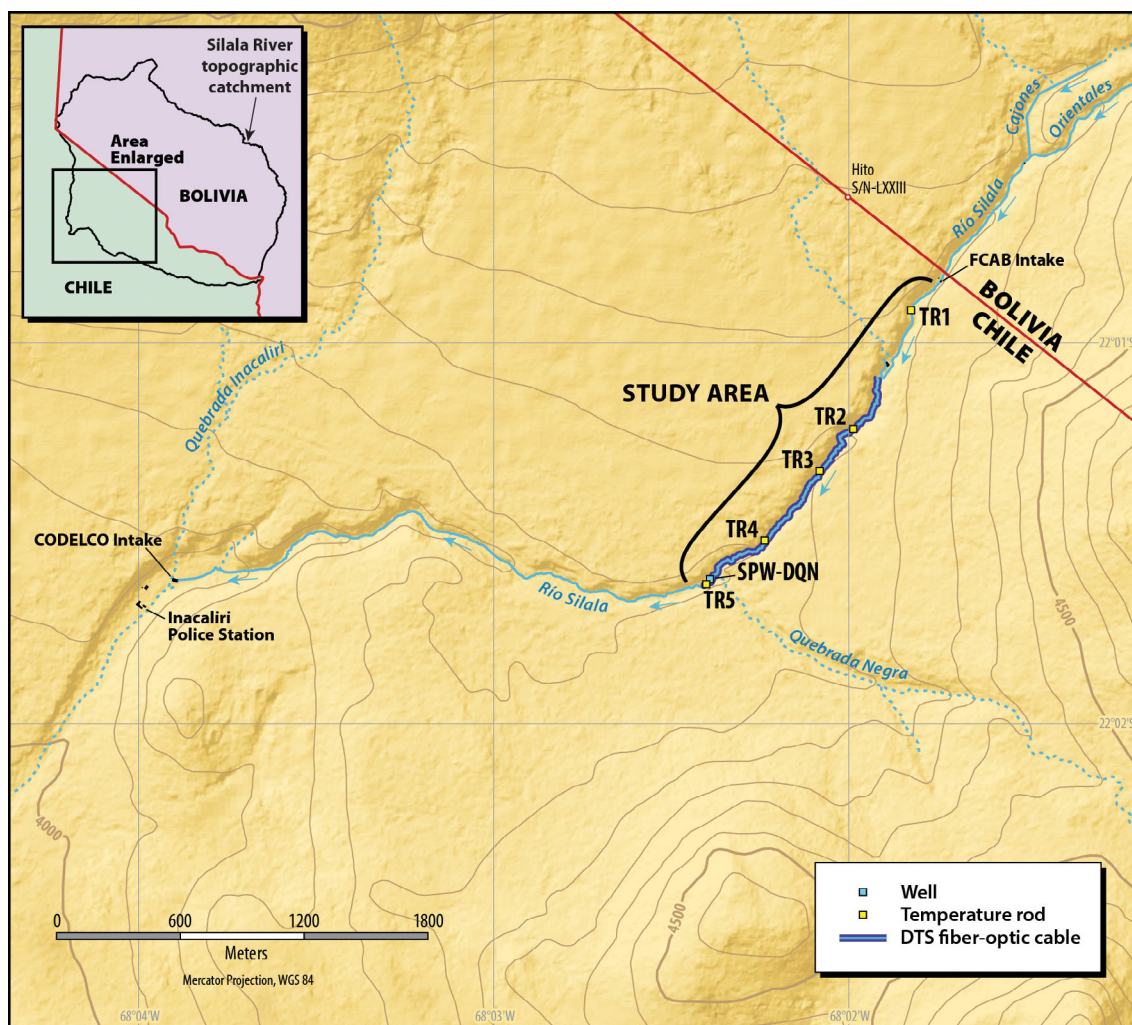


FIGURE 1 Location of the Silala River. The river reach where the interactions between groundwater and surface waters were investigated is highlighted. The figure shows the location of five temperature rods (TRs), the deployment of the distributed temperature sensing fiber-optic cable, the location of the SPW-DQN artesian well, and the location of the “The Antofagasta (Chili) and Bolivia Railway Company Ltd.” (FCAB) and CODELCO intakes (Suárez et al., 2017).

(Figure 1; CM, Vol. 1, 2017). The Silala river flows across the international border into Chilean territory at ~4277 m ASL (Alcayaga, 2017) and thence downstream towards Inacaliri. The nature of the Silala river as an international watercourse was not disputed until 1999, when Bolivia claimed its waters as exclusively Bolivian (CM, Vol. 1, 2017, pp. 11–15). A legal dispute between Chile and Bolivia over the status and use of the waters of the Silala river arose in 2016 due to Bolivia’s contention that the Silala River is not a transboundary watercourse (CM, Vol. 1, 2017). Chile requested that the International Court of Justice (ICJ) declare that the Silala river system is an international watercourse whose use by Chile and Bolivia is governed by customary international law (CM, Vol. 1, 2017, pp. 11–15).

The United Nations Watercourse Convention (UNWC), which reflects customary international law, establishes that (Rieu-Clarke et al., 2012, p. 75): “(a) ‘water course’ means a system of surface waters and groundwaters constituting by virtue of their physical relationship a unitary whole and normally flowing into a common terminus; (b) ‘International watercourse’ means a watercourse, parts of which are situated in different states.” Therefore, to support this legal dispute it was critical to provide scientific evidence concerning these aspects. Alcayaga (2017) clearly demonstrated that the Silala river flows from Bolivia into Chile following the natural topographic gradients, and Latorre and Frugone (2017) established that the Silala River has flowed through its current cross-border ravine for more than 8000 years. Building on improved geological mapping (Blanco & Polanco, 2024), Arcadis (2017) performed hydrogeological investigations in the study area to better understand the multi-aquifer system, including drilling, well testing and geophysical surveys (Gómez et al., 2024). Additional evidence was required to ascertain whether the surface waters and groundwaters constitute a unitary whole; so, in addition to geochemistry and isotope investigations (Herrera & Aravena, 2017),

the physical interactions between the river and the aquifer beneath it were further explored (Suárez et al., 2017), and are reviewed here. The results were also used in the development and calibration of a groundwater model of the basin (Taylor & Peach, 2024). Because of difficulties in accessing Bolivian territory, the study was undertaken in the river reach immediately downstream of the international border.

River–aquifer interactions have been extensively studied using traditional methods (Kalbus et al., 2006), but in recent decades the use of heat as an environmental tracer has flourished (Hatch et al., 2006; Irvine et al., 2017). Many methods have used streambed temperature time series to determine the magnitude and direction of flow exchange between rivers and aquifers (Hatch et al., 2006; Healy & Ronan, 1996; Keery et al., 2007; Luce et al., 2013; McAliley et al., 2022; McCallum et al., 2012; Munz et al., 2017; van Kampen et al., 2022). Also, thermal anomalies and energy-based models have used stream temperatures to investigate river–aquifer interactions (Hare et al., 2015; Henderson et al., 2009; Westhoff et al., 2007). In the following sections, both approaches are included: (a) the use of streambed temperature time series to estimate river–aquifer exchange fluxes; and (b) the use of innovative fiber-optic distributed temperature sensing (FO-DTS) methods to measure river temperatures, which allow thermal anomalies in the river to be related to inflows such as from groundwater springs. This article thus reviews the scientific work in which heat was used as a natural tracer to demonstrate that the Silala River is indeed a system of surface waters and groundwaters that interact as they flow into a common terminus, hence supporting the legal dispute over the status and use of the waters of the Silala river.

2 | THERMAL METHODS USED TO INVESTIGATE RIVER–AQUIFER INTERACTIONS IN THE SILALA RIVER BASIN

The two thermal methods used to investigate river–aquifer interactions in the Silala River catchment are complementary. Streambed temperature time series can be used to determine the magnitude of upwelling or downwelling fluxes through the streambed, whereas FO-DTS methods are typically used to detect groundwater inflow locations. This section describes the two methods, as well as their limitations for the investigation of river–aquifer interactions.

2.1 | Use of streambed temperature time series to estimate river–aquifer interactions

The temperatures in the upper part of a streambed, that is, in the first few meters, can be characterized using a vertical semi-infinite half-space domain. If the thermal conditions in this domain are assumed to be governed by a one-dimensional conduction–advection equation, the following heat transport equation can be used to describe the thermal dynamics (modified from Stallman, 1965):

$$C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[\lambda \frac{\partial T}{\partial z} \right] - C_w q \frac{\partial T}{\partial z} \quad (1)$$

where T is temperature; t is time; z is the vertical direction; C_p and C_w are the volumetric heat capacity of the saturated soil and water, respectively; q is the water flux in the vertical direction; and λ is the effective thermal conductivity of the saturated soil (Jury & Horton, 2004). Equation (1) implies that if the thermal spatiotemporal evolution in the sediments near the river is known, q can be inferred.

Different approaches have been used to determine q from Equation (1) when temperature time series at different depths in the sediments below a river are available (Hatch et al., 2006; Keery et al., 2007; Lautz, 2010; Luce et al., 2013; McCallum et al., 2012; Shanafield et al., 2011; Vandersteen et al., 2015). Gordon et al. (2012) developed the Vertical Fluid [Heat] Transport Solver (VFLUX) software, which was applied in the Chilean reach of the Silala River. This software was employed to estimate water fluxes between a pair of sensors within a temperature rod (TR) using signal analysis (Suárez et al., 2017). As shown in Figure 2, as depth increases, the thermal amplitude is reduced (ΔA) and a phase change ($\Delta \phi$) occurs. The changes in ΔA and $\Delta \phi$ are controlled by sediment thermal properties and water flux; consequently, they can be used to estimate q . In VFLUX, dynamic harmonic regression (DHR) is used to extract ΔA and $\Delta \phi$ from the temperature time series, and the water exchange between the river and the groundwater can be determined using methods based on: (i) amplitude ratio; (ii) phase change; and (iii) combined phase change/amplitude ratio. VFLUX has a total of six different analytical solutions that can be used, based on previous methods (Gordon et al., 2012; Irvine et al., 2017).

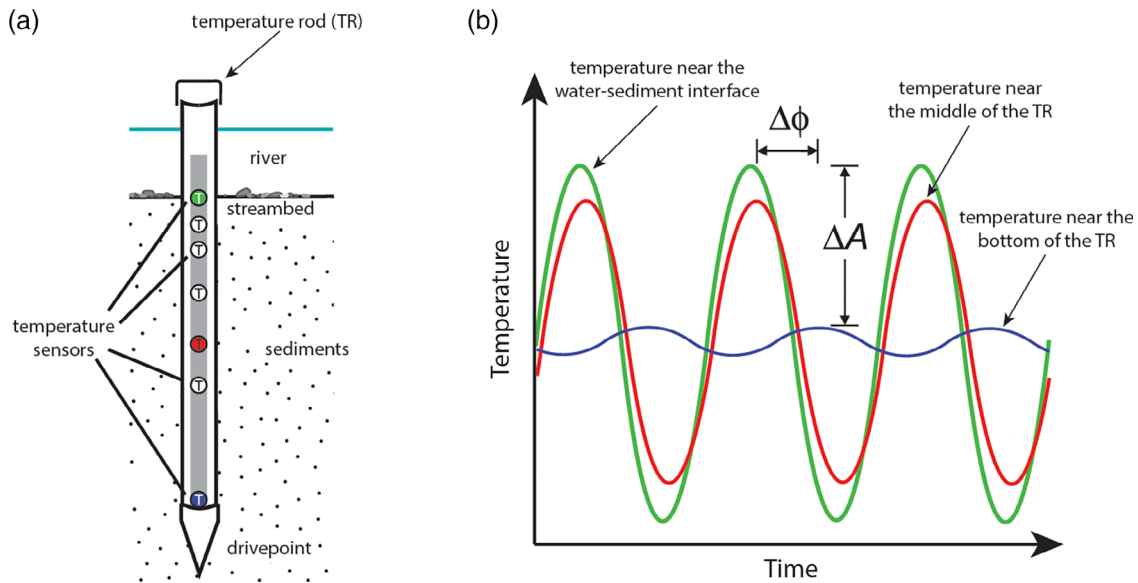


FIGURE 2 Diagram illustrating the upper (shallowest) part of a streambed and the thermal observations that are used to determine the water exchange between groundwater and surface water using thermal methods: (a) representation of a temperature rod (TR) used to collect temperature time series at different depths below the water–sediment interface; and (b) representation of the natural diurnal thermal signals collected at different depths in the TR. The temperature time series shows a reduction in amplitude (ΔA) and a shift in phase ($\Delta\phi$) with greater depth. ΔA and $\Delta\phi$ are used to determine the water exchange between groundwater and surface waters (modified from Hatch et al., 2006).

For the Silala river, the Hatch et al. (2006) amplitude method was chosen to determine the water exchange between groundwater and surface water (Suárez et al., 2017), as this method has shown good agreement with independent measured water fluxes for both downwelling and upwelling conditions (e.g., Briggs et al., 2014). The water fluxes are estimated according to Hatch et al. (2006):

$$q = \frac{C_p}{C_w} \left(\frac{2\lambda}{\Delta z C_p} \ln A_r + \sqrt{\frac{\alpha + v^2}{2}} \right) \quad (2)$$

$$\alpha = \sqrt{v^4 + \left(\frac{8\pi\lambda}{PC_p} \right)^2} \quad (3)$$

where A_r is the ratio of amplitude variations between pairs of temperature measurement points at different depths separated by Δz ; v is the velocity of the thermal front; and P is the period of temperature fluctuations. The thermal conductivity, heat capacity, thermal dispersivity, and porosity are needed to determine the water flux in the vertical direction (q).

The main limitations of this approach, when used to investigate river–aquifer interactions, are: (i) it uses an analytical solution valid for steady and uniform fluid velocity parallel with the vertical direction; and (ii) it assumes that the thermal properties of the porous media and the fluid are constant in space and time.

2.2 | Fiber-optic distributed temperature sensing (FO-DTS) theory and its use for detecting groundwater inflow locations

Fiber-optic distributed temperature sensing (DTS) systems utilize Raman spectra scattering to estimate temperature along the length of the fiber, achieving temperature resolutions as small as ± 0.01 °C, and spatial and temporal resolutions of 0.25–2 m and 1–300 s, respectively (Hausner et al., 2011; Lillo et al., 2022; Selker et al., 2006). This approach

delivers temperature measurements over space and time, $T(z, t)$, using the ratio of the anti-Stokes ($I_S(z)$) and Stokes ($I_{as}(z)$) Raman signals at the point of scattering, z , at time t (Van de Giesen et al., 2012):

$$T(z, t) = \frac{\gamma}{\ln\left(\frac{I_S(z, t)}{I_{as}(z, t)}\right) + C(t) - \int_0^z \Delta\alpha(z) dz} \quad (4)$$

where γ is the shift in energy between a photon at the wavelength of the incident light and the scattered Raman photon, C is a calibration parameter that includes properties of the incident laser and the FO-DTS instrument itself, and $\Delta\alpha$ is the differential attenuation between the anti-Stokes and Stokes signals within the fiber. The distance of the point of light scattering, z , is calculated by time-domain reflectometry using the speed of light in the glass fiber (Suárez et al., 2011). For successful calibration of the temperatures along the cable, at least three reference sections in different locations must be available (Hausner et al., 2011; Lillo et al., 2022; Van de Giesen et al., 2012). For a more detailed description of FO-DTS theory, the reader is referred to Dakin et al. (1985), Hausner et al. (2011), Rogers (1999), Suárez et al. (2011), and Van de Giesen et al. (2012).

FO-DTS has been used to investigate river–aquifer interactions by measuring the temperatures at the water–sediment interface (Selker et al., 2006; Westhoff et al., 2007). By observing these temperatures along cables of ~ 1 –4 km, and with spatial resolution of 0.5 m, this technique allows detection of localized groundwater inputs that can otherwise easily be missed. The main disadvantage of this approach is that it can be only used to infer groundwater inflow locations that result in significant thermal anomalies in the river temperatures. Therefore, small groundwater inputs may not be identified. To detect water flowing from the river to the aquifer, fiber-optic cables could be buried and heated, although that requires more effort to deploy the cables, as well as more energy required to heat the cables, which can be a major restriction in remote locations, such as the Silala River basin.

3 | FIELD METHODS USED IN THE SILALA RIVER CATCHMENT TO INVESTIGATE RIVER–AQUIFER INTERACTIONS

3.1 | Temperature rods (TRs) and piezometers, and thermal properties of the streambed sediments

Five TRs, each with a corresponding pair of piezometers, were installed between September and November 2016 along a 2 km reach of the Silala River in Chile (Figure 1). Each TR is similar to those designed by Naranjo and Turcotte (2015) and had ~ 6 –7 temperature sensors along it (DS1922L iButtons, Maxim Integrated, San Jose, CA; with an accuracy of $\pm 0.5^\circ\text{C}$ and a resolution of $\pm 0.0625^\circ\text{C}$). The thermal data were collected at 1-h time intervals. The first piezometer was used to measure water levels in the river, whereas the second piezometer measured water levels in the aquifer beneath the river. These water levels were measured at time intervals of 15 min using pressure transducers (HOBO Water Level [4 m] Data Logger, Onset Computer Corporation, Bourne, MA; with a typical error of 4 mm, a maximum error of 8 mm, and a resolution of 1.4 mm). These piezometers allowed determination of the hydraulic gradient between the river and the fluvial aquifer at each TR, which indicates flow direction. For the details of this deployment, the reader is referred to Suárez et al. (2017), whereas Arcadis (2017) describe the different aquifers in the basin.

The thermal properties of the sediments were estimated using the dual probe method with transient heating, which allows λ and C_p to be determined by solving the radial heat flow equation (Bristow et al., 1994). This method was implemented in the field using a thermal properties analyzer (KD2 Pro/SH-1, Decagon Devices, Pullman, WA), where typically three or more measurements were made at each TR. Table 1 shows the field measurements of thermal conductivity and heat capacity found by Suárez et al. (2017), which are input parameters for the Hatch et al. (2006) method. The average thermal conductivity values varied between 0.995 and $1.990\text{ W m}^{-1}\text{ K}^{-1}$. These values are smaller than thermal conductivities found in similar studies, for example, Gordon et al. (2012) utilized thermal conductivity values of 1.3 and $2.26\text{ W m}^{-1}\text{ K}^{-1}$. The measured volumetric heat capacity ranged between 2.83 and $3.61\text{ MJ m}^{-3}\text{ K}^{-1}$, which is in agreement with other investigations (Briggs et al., 2016).

TABLE 1 Field measurements of thermal conductivity (λ) and heat capacity (C_p) at each TR (average value $\pm$ SD; Suárez et al., 2017)

Temperature rod (TR)	TR1 ^a	TR2 ^a	TR3 ^b	TR4 ^c	TR5 ^b
Thermal conductivity, λ (W m ⁻¹ K ⁻¹)	1.632 $\pm$ 0.963	0.995 $\pm$ 0.370	1.990 $\pm$ 1.600	1.241 $\pm$ 0.436	1.940 $\pm$ 1.45
Heat capacity, C_p (MJ m ⁻³ K ⁻¹)	3.031 $\pm$ 0.408	2.836 $\pm$ 0.587	2.901 $\pm$ 1.329	2.903 $\pm$ 0.950	3.613 $\pm$ 1.268

^aThree measurements.^bFour measurements.^cTwo measurements.

3.2 | FO-DTS field deployment—Calibration metrics and filtering process to obtain river temperatures

With the aim of identifying gaining reaches along the Silala river, Suárez et al. (2017) performed double-ended FO-DTS measurements between September 30, 2016 and January 13, 2017. They used an XT-DTS (Silixa, Hertfordshire, UK) instrument and a 2-km long fiber-optic cable (BRUsens temperature 85 °C, Brugg Kabel AG, Windisch, Switzerland). The spatial sampling resolution was $\sim$ 0.25 m and the temporal integration was 5 min. The start of the cable deployment was located approximately 2 km downstream of the international border between Chile and Bolivia (Figure 1), and the cable was installed in the river over approximately 1.3 km, going upstream, passing through seven culverts.

Figure 3a illustrates the FO-DTS deployment, and examples of the raw data and temperature traces are shown in Figure 3b, c. The cable location at the most upstream point underwater was used as the origin for the distance along the cable (Suárez et al., 2017). At some locations along the river, it was not possible to avoid obstructions (e.g., due to the presence of pipes, weirs, and in some places very dense vegetation) so several sections of the cable were exposed to air (Figure 3c). In some of these, the cable was exposed to direct sun while in others it was shaded by vegetation. Figure 3b demonstrates that: (i) the cable was not damaged during the deployment, as no step losses are observed in the raw Raman signals (Hausner et al., 2011; Hausner & Kobs, 2016), and (ii) the level of the Raman intensities is sufficient to allow collection of high-quality data (>500 arbitrary units at the end of the cable).

Double-ended measurements were carried out and used reference sections X_1 and X_3 to perform a dynamic calibration of the temperatures along the entire cable (Figure 3a, c). The calibration metrics used by Hausner et al. (2011), mean bias (MB) and root mean squared error ($RMSE$), were employed to report the thermal resolution of this FO-DTS deployment. To account for the natural variations of the system, Suárez et al. (2017) selected two sets of time records for improved thermal resolution: 4 h centered around the maximum temperature recorded in the water during each day (referred to as “day”), and 4 h prior to the lowest recorded temperature in the water each day (hours with no influence of solar radiation, referred to as “night”). The calibration metrics for the 5250 temperature traces (50 traces per day, collected on 105 days) averaged over the day and the night are reported in Table 2, together with the metrics for the mean temperature (104 complete days, 29,701 traces). Table 2 shows that the thermal resolution (reported as the $RMSE$) of the system for the day and night traces was 0.14 and 0.11 °C, respectively, in the worst case. For the day and night traces, $RMSE$'s as small as 0.02 and 0.01 °C, respectively, were observed at reference sections that were not used to calibrate the temperatures along the cable. Because during the day solar radiation absorption along the cable is likely to affect the thermal measurements, it is better to use the temperatures measured during the night for studying river–aquifer interactions. Therefore, as the fiber-optic cable was not damaged during the deployment, and the night traces averaged over 105 days have a thermal resolution as small as 0.11 °C, any observed thermal anomaly larger than 0.11 °C must come from a physical process, such as water entering the river, that occurs consistently throughout the monitoring period.

Before investigating river–aquifer interactions using the thermal data from the river, measurements that do not correspond to water temperature were removed (Sarabia, 2017). Data filtering considered that: (i) atmospheric temperature is measured in the cable exposed to air, and is subject to daily oscillations that results in large temporal standard deviation; (ii) water temperature is measured in the cable submerged in the river, and is subject to reduced temperature oscillations compared with the sections exposed to air (Stonestrom & Constantz, 2003). Also, river temperatures are higher than 0 °C unless the water freezes; and (iii) the cable sections that measure the temperature of groundwater discharge are subject to lower thermal oscillations than those of the river waters, as groundwater has a relatively constant temperature (Stonestrom & Constantz, 2003). These thermal features allowed removal of air temperatures from the collected

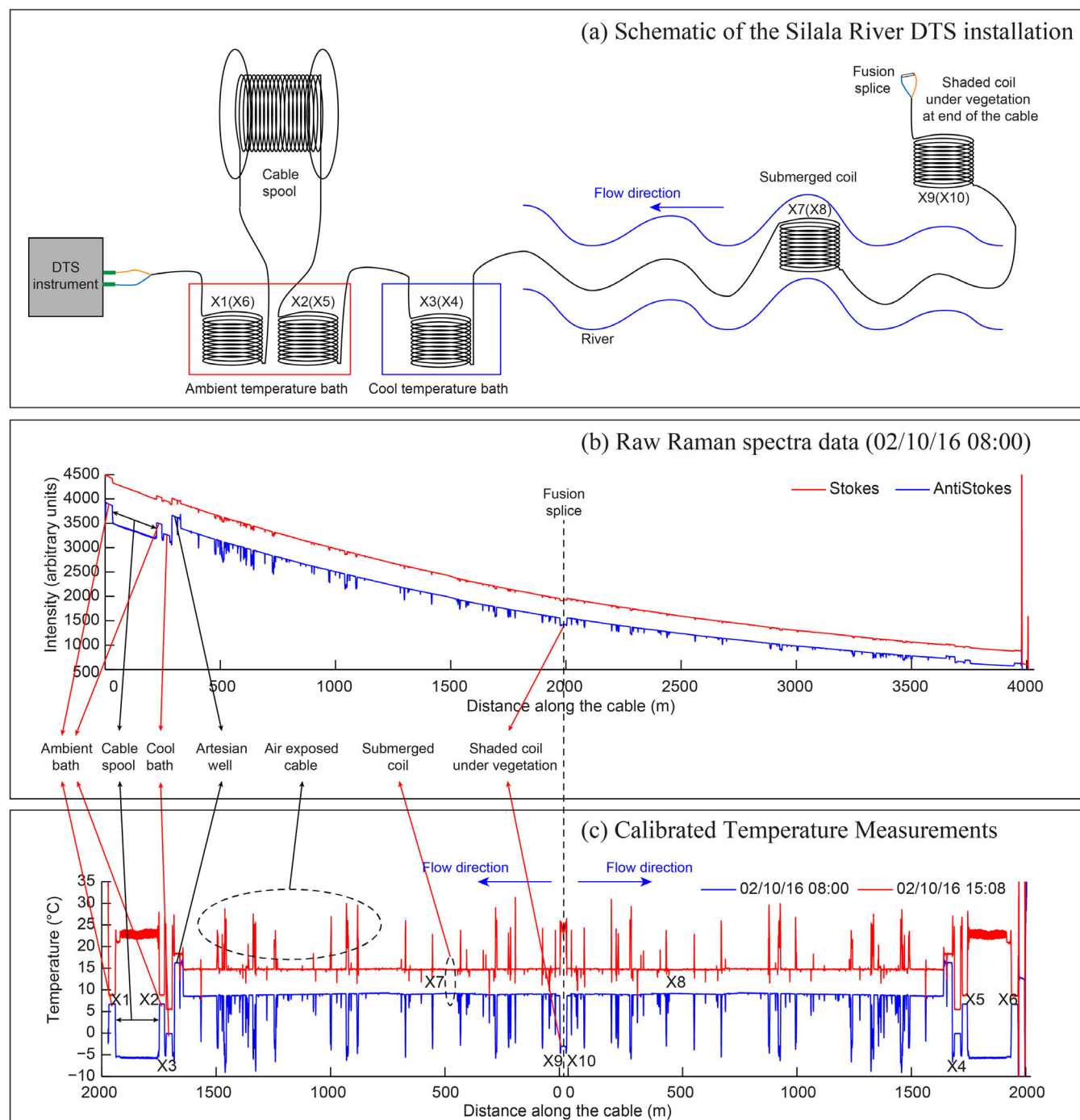


FIGURE 3 Silala river FO-DTS installation: (a) schematic of the field installation; (b) raw Raman spectra data (recorded by the FO-DTS instrument in arbitrary units linearly related to the power of the scattered signals) and reference locations for a 5-min temperature trace sample (October 2, 2016—8:00). In this subfigure, the distance along the cable uses the instrument position as the coordinate's origin; (c) temperatures and the locations of reference zones (X_i) for 5-min temperature trace samples at daily minimum (October 2, 2016—8:00) and maximum temperatures (October 2, 2016—15:08). In this subfigure, the distance along the cable uses the most upstream point underwater as the coordinate's origin. Modified from Suárez et al. (2017).

data, resulting in a thermal record of the water temperatures along the river reach. Figure 4 presents the mean thermal profile for the day and night, as well as the temperature standard deviation profile, and their respective underlying trends, which represent the actual values of temperature and standard deviation in the river waters at each location. These trends were determined using the LOWESS algorithm (Cleveland, 1979).

TABLE 2 Calibration metrics for the Silala river FO-DTS deployment using temperature traces obtained using all the temperature records of complete days (“Mean”), using an average from 4 h before and up to the time when the minimum temperatures were observed (“Night”), and using an average of 4 h of data centered at solar noon (“Day”)

Location	Mean bias (MB; °C)			Root mean squared error (RMSE; °C)		
	Mean	Night	Day	Mean	Night	Day
X_1	0.08	0.09	0.06	0.09	0.10	0.07
X_2	0.10	0.11	0.07	0.10	0.11	0.07
X_3	0.00	0.01	−0.02	<0.01	0.02	0.02
X_4	−0.54	0.00	−0.10	0.05	0.01	0.10
X_5	0.035	0.09	−0.02	0.03	0.09	0.03
X_6	−0.07	−0.01	−0.14	0.08	0.04	0.14

Note: Figure 3 depicts the locations (X_i) of the reference zones used for calibration and validation (Suárez et al., 2017).

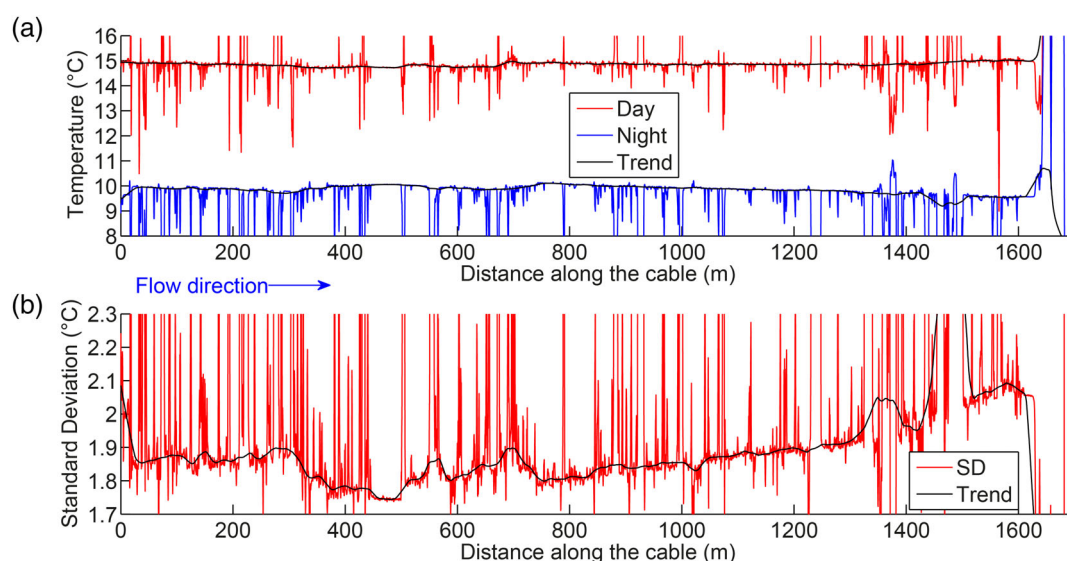


FIGURE 4 (a) Mean temperature profile obtained with FO-DTS for the day and the night and respective underlying trend. (b) Temperature SD profile and trend (Suárez et al., 2017).

4 | RIVER-AQUIFER WATER FLUX EXCHANGE, AND DETECTION OF GROUNDWATER INPUTS

4.1 | Estimation of water flux exchange from temperature time series

For brevity, only the results obtained at TR1 are presented and discussed here in detail, as they are representative of most of the investigated TRs. The observed temperature time series at different depths within the streambed and the flow exchange between surface waters and groundwater are presented for all the TRs installed in the study zone. For further details, the reader is referred to Suárez et al. (2017).

4.1.1 | TR1: Water exchange fluxes and uncertainty analysis

The temperature time series at different depths at TR1 are shown in Figure 5a. These show variable daily fluctuations between 10 and 16 °C at 0.5 cm depth. All the temperature time series increase their daily minimum temperature as the austral summer (which corresponds to the wet season) approaches. Figure 5a also shows that the diurnal temperature signal amplitude is attenuated below 42.3 cm depth. Figure 5b presents the river stage and groundwater level

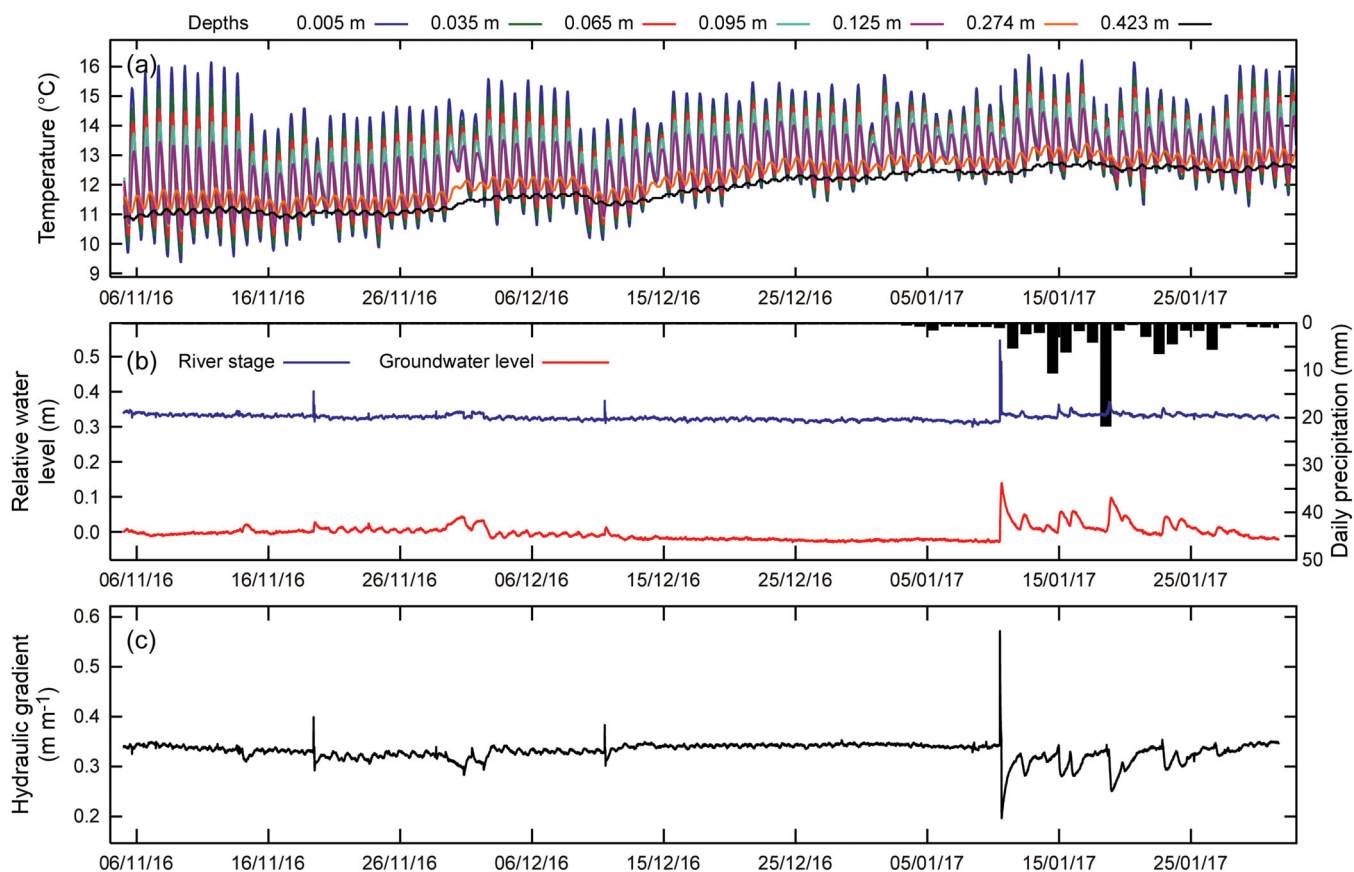


FIGURE 5 Thermal and hydraulic data collected at TR1: (a) temperature time series at different depths; (b) river stage, groundwater level and daily precipitation; and (c) hydraulic gradient between the river and the aquifer (Suárez et al., 2017).

measured by the pressure transducers at TR1, and the precipitation measured at the Inacaliri meteorological station (see Figure 1) (Muñoz et al., 2017). The water levels are relative to the groundwater level measured at 0:00 on November 5, 2016. The river stage displays diurnal cycles, which can in principle be used to understand some of the dominant processes affecting the water balance of a river basin. For instance, evapotranspiration, infiltration in losing reaches, diurnal cycles in precipitation and snowmelt are physical mechanisms that can explain diurnal cycles (Lundquist & Cayan, 2002), such as those reported by Suárez et al. (2017). However, Suárez et al. (2017) only had approximately 3 months of water level data, which was considered inadequate to determine the mechanisms that explain these diurnal cycles. Figure 5b shows that the effect of precipitation is more noticeable for groundwater levels than for the river stage. The abrupt change in water levels observed on January 10, 2017 was due to a sediment transport experiment where the “The Antofagasta (Chili) and Bolivia Railway Company Ltd.” (FCAB) stopped the diversion of water into the FCAB intake (Mao, 2017). This caused a spike in the river flow and a groundwater response that is bigger than the effect of precipitation events on groundwater levels (this can be observed when comparing the increase in groundwater levels between the day of the sediment transport experiment and the day when the largest precipitation occurred). These data suggest that groundwater recharge occurs rapidly and that water leaks through the riverbed, flowing towards the fluvial aquifer. Figure 5b also shows that there are some perturbations in the water levels when there is no precipitation. The perturbations observed between November 28, 2016 and December 2, 2016 are due to pump tests carried out in the well PW-BO, which is located near the Chile-Bolivia international border, where the water pumped was discharged into the river (at a location downstream of the well but upstream of TR1) (Arcadis, 2017; Gómez et al., 2024). The other perturbations observed in the water levels cannot be explained by the field data that were collected; they might be related to the operation of the FCAB water intake or due to natural processes. Figure 5c presents the hydraulic gradient at TR1. The average hydraulic gradient is approximately 0.33 m m⁻¹ with a corresponding pressure head difference of 0.33 m between the river and the aquifer, implying that the water exchange must be from the river to the aquifer.

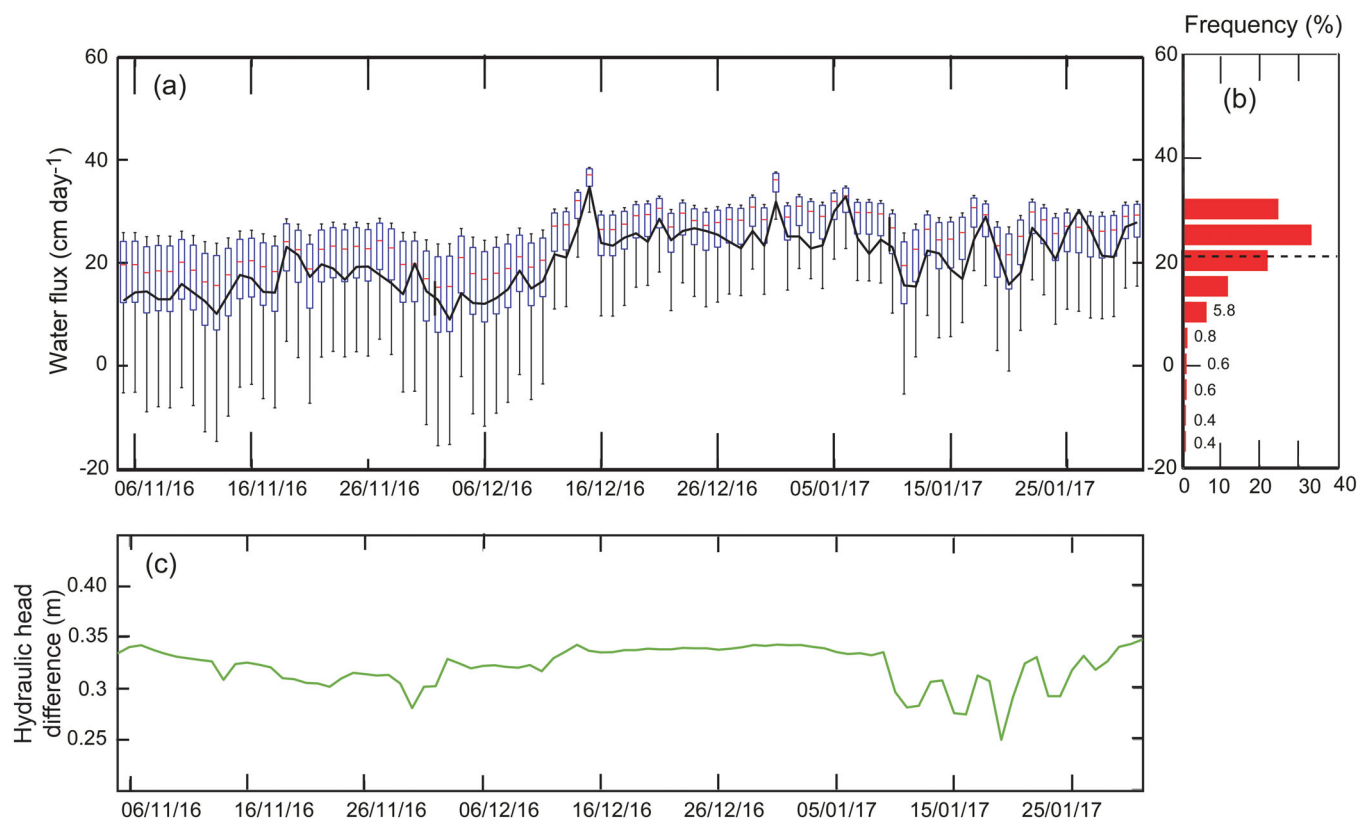


FIGURE 6 (a) Daily mean water fluxes (cm³ cm⁻² day⁻¹) between the river and the aquifer at 0.02 m depth, estimated with the average thermal properties at TR1 (continuous black line) and with Monte Carlo simulations (boxplot). Positive values correspond to downwelling conditions and negative values correspond to upwelling conditions. (b) Histogram of the mean water flux of each simulation throughout all the studied period obtained with the Monte Carlo simulations (red columns). The dashed line corresponds to the mean water flux estimated using the average thermal properties. The numbers above the red columns are the frequency percentages below 10%. (c) Daily hydraulic head difference between the river and the aquifer. Positive values are related to water flowing from the river to the aquifer (Suárez et al., 2017).

BOX 1 Monte Carlo experiments allow investigating water fluxes uncertainty

For the Silala River, the statistical confidence intervals of the water fluxes were determined with the Monte Carlo method, which is a valuable approach to investigate the uncertainty of the estimated water fluxes. The input parameters of each realization were thermal conductivity, volumetric heat capacity, and porosity. Hence, Monte Carlo simulations did not include uncertainty related to violations of model assumptions. The results of the uncertainty analysis were reported using boxplots (e.g., see Figure 6a), in which the lower edge of the box represents the first quartile of the data (25th percentile), the horizontal red line inside the box corresponds to the median of the data (50th percentile), and the upper edge of the box is the third quartile of the data (75th percentile). In the boxplots, the plotted whiskers extend to the most extreme data value that is not an outlier. As visualized in Figure 6a, even when there is an intrinsic uncertainty in the input data of the time series analysis, the water fluxes are likely to be directed downwards (from the river towards the aquifer).

Figure 6a presents the daily mean water fluxes between the river and the aquifer estimated using the average thermal properties at TR1 (continuous black line) and with Monte Carlo simulations (see Box 1). The Monte Carlo simulations performed by Suárez et al. (2017) only accounted for expected uncertainty in model thermal input parameters; other sources of uncertainty that come from violations of model assumptions are not included in the analysis.

The water fluxes shown in Figure 6a were determined at 0.02 m depth using VFLUX. Figure 6a shows that the water fluxes estimated with the average thermal properties varied between 9 and 35 cm day⁻¹ during the monitoring period, with an average value of 21 cm day⁻¹ (dashed line in Figure 6b). The uncertainty analysis simulations reveal that when using heat as a tracer without considering the hydraulic gradient, the probability of having a water exchange from the river to the aquifer is 98%. Some increases in downward water flux observed in Figure 6a could be due to non-ideal input signals. Nonetheless, the flux values in the vicinity of these increases typically are greater than zero (i.e., downwards), so that probability is not expected to change. The uncertainty analysis yields water fluxes of 22 ± 6 cm day⁻¹ (mean ± SD of 921 simulations), with a median of 24 cm day⁻¹. Figure 6b shows the frequency distribution of the average fluxes obtained with the Monte Carlo simulations. The distribution is asymmetrical and the estimated water fluxes are significantly different from zero, which implies that river–aquifer interactions are occurring. Figure 6c shows the mean daily values of hydraulic head difference in the period of study. Overall, the direction of the water fluxes estimated using temperature time series agrees with the hydraulic gradient data. There is a good correlation between the timing of the variations in the water fluxes and of the fluctuations in the hydraulic head difference, although this timing shows a small shift on some days. The differences related to temporal fluctuations of water fluxes and hydraulic head difference are likely due to limitations of the analytical method used to quantify the water exchange between the river and the aquifer (Hatch et al., 2006; Vandersteen et al., 2015).

4.1.2 | Comparison of the measurements between the TRs and limitations of the temperature time series analysis

Figure 7 shows the temperature time series for all the TRs analyzed in this study during the period November 5, 2016 to January 31, 2017. The maximum mean daily amplitudes, measured near the water–sediment interface, were: 3.4 °C at TR1; 1.3 °C at TR2; 1.6 °C at TR3; 4 °C at TR4; and 1 °C at TR5. Locations with small diurnal temperature signal amplitude may also indicate groundwater input into the river not necessarily associated with water exchange occurring across the water–sediment interface. For instance, the small diurnal temperature signal amplitude observed at TR5 is the result of a warm groundwater discharge that comes from an artesian well (SPW-DQN). The waters of the SPW-DQN well enters the river approximately 30 m upstream TR5. Similar processes to those occurring at SPW-DQN can occur for the springs that discharge into the river, for example, the small diurnal temperature signal amplitude observed at TR2 and TR3 is likely the result of spring discharge from the walls of the ravine. Nonetheless, the results obtained by Suárez et al. (2017) show with a high probability that in the streambed water is flowing from the river towards the aquifer at these locations.

The water fluxes estimated using the mean thermal properties at each TR are presented in Figure 8. The Monte Carlo analysis (Box 1) yielded estimates of water fluxes with their uncertainty: 22 ± 6 cm day⁻¹ (mean ± SD) (TR1); 5 ± 7 cm day⁻¹ (TR2); -7 ± 21 cm day⁻¹ (TR3); 44 ± 7 cm day⁻¹ (TR4); and 3 ± 16 cm day⁻¹ (TR5). In general, all the estimated water fluxes are smaller than 60 cm day⁻¹. Assuming a river width of 1 m and a river reach of approximately 2 km, the estimated mean water flow discharged from the river into the aquifer through the riverbed is of 3.3 ± 2.7 L/s (Suárez et al., 2017). Therefore, the water exchange through the riverbed is relatively small compared with the river flow, which is of ~140 L/s at the Chile-Bolivia international border (Yáñez-Morroni et al., 2024a, 2024b). The magnitude of the fluxes found by Suárez et al. (2017) is of the same order as other investigations. For instance, Gordon et al. (2012) found positive fluxes of ~86 cm day⁻¹ at 15 and 25 cm depth. Also, the extinction depths are similar to those obtained by Briggs et al. (2012); between 30 and 60 cm, with amplitude signals over 0.2–1.2 °C.

The main limitation of the Hatch et al. (2006) amplitude method is that it uses an analytical solution valid for steady and uniform fluid velocity parallel with the vertical direction. Also, the analytical solution applies when the heat properties of the medium and fluid are constant in space and time, and when there is thermal equilibrium between the porous media and the fluid. Therefore, this method is not sensitive enough to detect abrupt changes that occur in the seepage rates or in the water flux direction. In many natural systems the seepage rates will vary slowly over days, weeks, months or even seasons and this limitation is not relevant (Hatch et al., 2006). Nonetheless, rapid changes can occur in natural systems as the result of human activities, which occurred in the Silala River during the period where river temperatures were measured (Arcadis, 2017; Gómez et al., 2024). Also, because the analytical method estimates a 2 h constant water flux (Gordon et al., 2012), it is not able to capture the variability that was observed in the hydraulic head data that were collected at 15 min time intervals. The use of the Hatch et al. (2006) amplitude method alone, that is, without using the phase shift as an additional input, may lead to error in flux estimations when field data are used to

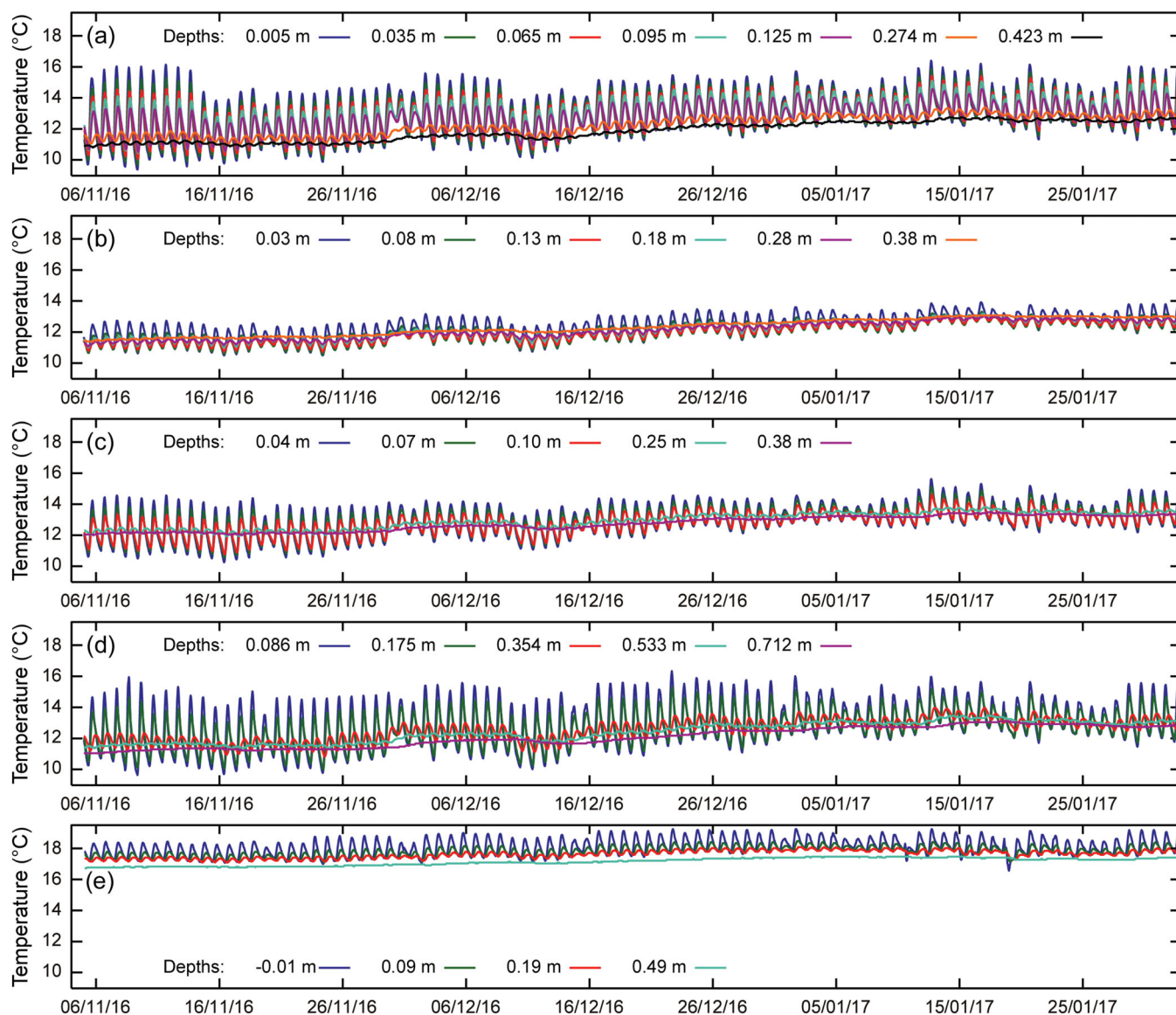


FIGURE 7 Temperature time series for all the TRs. (a) TR1; (b) TR2; (c) TR3; (d) TR4; and (e) TR5.

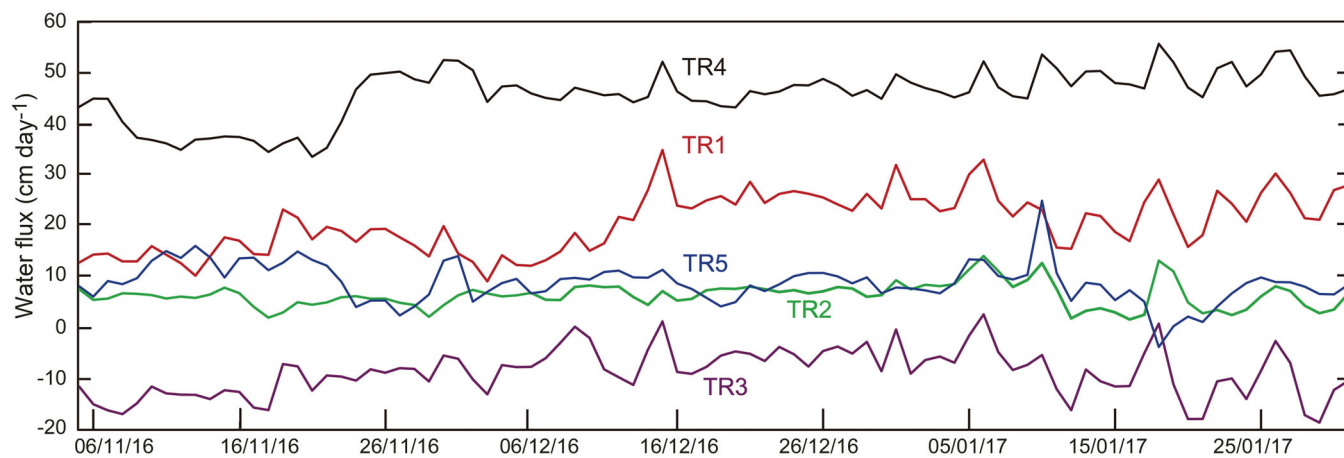


FIGURE 8 Comparison of the daily mean water fluxes between the river and the aquifer estimated using the Hatch et al. (2006) amplitude method and the average thermal properties at each TR.

populate the model or when three-dimensional flow structures appear in the seepage velocity field (Hatch et al., 2006; Vandersteen et al., 2015). The Hatch et al. (2006) amplitude method also assumes that the porous medium is homogeneous, which typically does not occur in natural systems; and the Silala riverbed is not the exception (Latorre & Frugone, 2017). This limitation increases the uncertainty in the estimation of the water fluxes when the porous medium is stratified. Finally, it is important to notice that the exchange flux estimates were obtained using approximately 3 months of data. It is likely that when considering longer time series, the magnitude of these estimates will change and must be further studied.

Despite all the limitations discussed above, as shown by the Monte Carlo simulations, in this study the water fluxes estimated using thermal methods are significantly different from zero in most of the TRs (Suárez et al., 2017). Therefore, the interactions between groundwater and surface waters are clearly demonstrated. These interactions were also observed using the hydraulic head difference that was measured with pressure transducers. In general terms, the results obtained by Suárez et al. (2017) suggest that water flows from the river to the aquifer through the streambed sediments. However, **since the observations are point-in-space measurements, it is possible that the flow is reversed in other locations of the river reach. In fact, in the hyporheic zone of mountain streams it is common to find both downwelling and upwelling zones** (Buffington & Tonina, 2009; Tonina & Buffington, 2009; Westhoff et al., 2007; Wroblicky et al., 1998).

4.2 | Detection of the locations of groundwater discharge to the river using FO-DTS

The day and night mean temperature profiles shown in Figure 4a were used to identify possible locations of groundwater discharge to the river (Sarabia, 2017; Suárez et al., 2017). Three types of possible river–aquifer interactions were detected in the Silala River:

4.2.1 | Type-1 (T1) river–aquifer interaction

Since groundwater in this basin has a higher temperature than the river, a step increase in the night temperature of at least 0.1 °C, with evident effects downstream, was identified as a location of groundwater discharge, named Type-1 (T1) river–aquifer interaction.

The T1 river–aquifer interactions could only be produced by local sources of heat present during the night since there is no direct radiation and any residual heat source from the sediments should also be negligible. Hence, the abrupt river temperature increases during the night can only be explained by warm groundwater discharges (from springs or from the streambed). The T1 temperature anomalies can also be seen in the daytime temperature trace at the same location. This type of groundwater input was also reported by Westhoff et al. (2007). They detected temperature changes as large as 2 °C at specific locations in a stream, and used an energy conservation dynamic model to demonstrate that those thermal anomalies could only be explained by groundwater inputs. In the case of Westhoff et al. (2007), the thermal anomalies were more clearly identified during the day because the groundwater inputs were cooler than the stream temperature, while in the Silala river the sources are generally warmer than surface waters and are easily identified during the night (Sarabia, 2017).

4.2.2 | Type-2 (T2) river–aquifer interaction

A location that exhibits a temperature peak during the night and a temperature valley during the day was identified as a location where a river–aquifer interaction occurs, and was referred to as Type-2 (T2) river–aquifer interactions. There are two plausible explanations for this temperature anomaly: (1) if the cable is measuring the temperature at the water–sediment interface, this thermal anomaly can only be explained by a location of groundwater upwelling; and (2) if the cable is buried and measures the temperatures of the riverbed sediments, this thermal anomaly can be explained by a location where water exchange between the river and the aquifer occurs (upwelling or downwelling conditions). Thermal conduction could also explain thermal anomalies in a buried fiber-optic cable, but Suárez et al. (2017) demonstrated that this was not the case for the study performed in the Silala River.

According to the scientific literature, there are two possible explanations for the thermal anomalies associated with the T2 river–aquifer interactions (Hatch et al., 2006; Krause et al., 2012; Selker et al., 2006; Westhoff et al., 2007).

The first is that the temperature anomaly is related to a localized groundwater discharge that has a temperature within the range of the river temperatures (Selker et al., 2006). In the case of the Silala River, the T2 groundwater discharges are not strong enough as to influence the temperature of the river water downstream. This thermal behavior has also been reported in other investigations (Krause et al., 2012). Therefore, in this situation the cable would be located just above a localized groundwater discharge of relatively low flow (compared with the river flow). To be able to detect this type of river-aquifer interaction using FO-DTS methods, the fiber-optic cable should be measuring the water-sediment interface. The second explanation applies when the cable is buried below the water-sediment interface (Krause et al., 2012). As described before, as depth increases, the thermal amplitude is reduced, and a phase change occurs. The amplitude reduction and phase change are associated with the thermal properties of the sediments and the water travel time through the sediments. Thus, they could also be used to quantify the magnitude and direction of the water exchange between the river and the aquifer. During a field trip carried out during January, 2017, Suárez et al. (2017) found that the T2 river-aquifer interactions that were identified are buried sections of cable where it is more likely that water flows from the river to the aquifer, which is consistent with the TR findings presented above.

4.2.3 | Type-3 (T3) river-aquifer interaction

Groundwater coming from a deeper aquifer than the fluvial aquifer was referred to as the Type-3 (T3) river-aquifer interaction. In the Silala River, this situation corresponds to the SPW-DQN artesian well (see Box 2), which produces a similar effect to the Type-1 groundwater discharge, but its effect on the thermal dynamics of the river was important during the measurement period. Due to this groundwater input, the downstream river temperature increased by more than 5 °C during the day and it exhibited small diel thermal variability compared with the upstream thermal conditions of the river.

The T3 river-aquifer interaction is a point-source groundwater discharge that results from an artesian well discharging into the river (Arcadis, 2017; Gómez et al., 2024). Interestingly, the location of the discharge of this well into the river was not known when the fiber-optic cable was deployed, but it was discovered with precision thanks to the high spatial resolution (~0.5 m) of the FO-DTS measurements (as described in Box 2).

4.2.4 | Conceptual model of river-aquifer interactions

Using the FO-DTS data, Suárez et al. (2017) proposed a conceptual model that indicates the locations of groundwater discharge that were identified (Figure 9). These locations are presented as the distance along the fiber-optic cable, but

BOX 2 Fiber-optic distributed temperature sensing data provides a way to find thermal anomalies at disparate spatial scales—the story of the SPW-DQN artesian well

The SPW-DQN well is an artesian well located at the confluence of the Silala river and Quebrada Negra. It is located approximately 10 m from the river in the riparian wetland, and discharges its waters into the Silala River and into the fluvial aquifer beneath the river. SPW-DQN was drilled for exploratory purposes in 1995 and reached a depth of approximately 70 m (Arcadis, 2017; Gómez et al., 2024). At that time, the Chilean National Directorate of Waters did not authorize the water rights of this well and thus it was sealed. Nonetheless, due to its artesian nature, the pressure of the water forced it to flow to the ground surface, which meant that in practice groundwater was still emerging at this location with approximate flow and temperature of 90 L/s and 20 °C, respectively. Upstream of SPW-DQN, the flow of the Silala river flow and its temperature were approximately 50 L/s and 10 °C, respectively. Therefore, a large thermal contrast occurred at this location, and the large thermal mass provided by SPW-DQN changed the downstream thermal dynamics of the river: increased water temperatures and reduced diel variability compared with upstream conditions. This water discharge was not known prior to the fiber-optic distributed temperature sensing (FO-DTS) deployment. FO-DTS allowed finding the well's discharge location precisely: with a spatial resolution of 0.5 m in a total reach of more than 1 km.

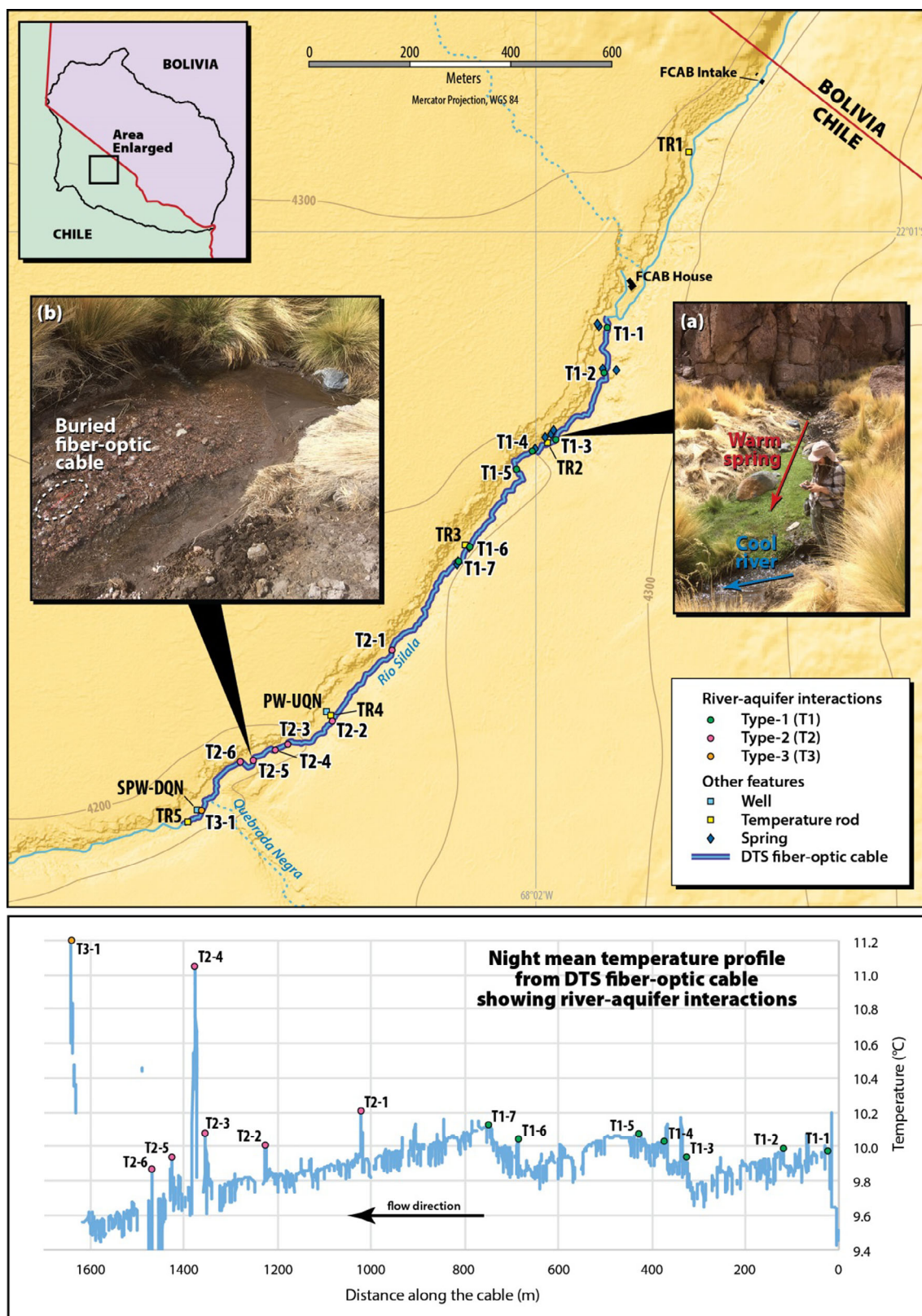


FIGURE 9 Conceptual model of the locations of main groundwater contributions detected using FO-DTS methods. (a) Photograph depicting one of the springs entering the Silala river. (b) Photograph depicting a zone where the fiber-optic cable was buried (Suárez et al., 2017).

also are depicted with geographical locations and physical references (this is the reason why the FO-DTS temperature profile is inverted in Figure 9). Green circles represent a consistent increase in temperature with effects downstream (T1 river-aquifer interaction). The circles in magenta represent locations with an evident trend to a stable temperature when comparing day and night temperatures, and a considerably higher temperature during night, but with no apparent effect on temperatures downstream (T2 river-aquifer interaction). The orange circle represents a relevant and consistent increase in temperature that affects downstream conditions (T3 river-aquifer interaction), which is caused by the artesian well SPW-DQN. The groundwater discharge locations that were identified using FO-DTS methods are coherent with the field observations of springs, as shown in the photograph of Figure 9a. Typically, these springs discharge water from the ravine walls at elevations higher than the river stage. The location of the springs that have enough thermal mass to change the river temperature by more than $\sim 0.1^\circ\text{C}$ is also indicated in Figure 9.

5 | SUMMARY AND CONCLUSIONS

Thermal methods were applied in the Chilean reach of the Silala River to investigate river-aquifer interactions. The river reach began at the Chile-Bolivia international border and finished approximately 2 km downstream of the border, at the confluence with Quebrada Negra.

Time series analysis suggests that the river feeds the fluvial aquifer located below the streambed throughout the investigated river reach (losing river). Water fluxes as large as 50 cm day^{-1} were estimated using the Hatch et al. (2006) amplitude method. The water flux direction estimated with this method is consistent with the measured vertical hydraulic gradient, which also indicates that water flows from the river to the aquifer. Nonetheless, in some locations the hydraulic gradient was small, implying that changes in river stage or groundwater level could reverse the direction of the water fluxes. An uncertainty analysis, which considered variability in the input parameters of the thermal model used, revealed that the water exchange between groundwater and surface waters is different from zero for most of the TRs. Therefore, the river-aquifer interactions are clearly demonstrated. Nonetheless, it is important to recall that the water exchanges between the river and the aquifer quantified using temperature time series are point-in-space measurements and consequently, they do not prove that the same behavior occurs along the entire river.

FO-DTS methods were used to investigate river-aquifer interactions along a river reach that began at $\sim 750\text{ m}$ from the Chile-Bolivia international border and finished $\sim 2\text{ km}$ downstream of the border. Along this reach, temperature was measured with a sampling resolution of 0.25 m and an integration time of 5 min . The collected temperature data were separated into mean profiles taken during the day and the night, and thermal anomalies larger than 0.11°C were used to detect locations of warm groundwater discharge. Several locations of groundwater discharge were identified, but the most distinct groundwater input was the T3 river-aquifer interaction. This warm discharge corresponds to the SPW-DQN artesian well, which heavily influences the thermal behavior of the river downstream, producing an increase of $\sim 5^\circ\text{C}$ in the mean water temperature during the entire day, and up to $\sim 9^\circ\text{C}$ at specific times during the night. The T1 river-aquifer interactions were the second most important groundwater discharges along the study reach. The thermal anomalies caused by the T1 inputs are clearly present during the whole measuring period and were consistent in their locations, increasing the downstream temperature of the river. These locations are highly correlated with the positions of the springs that discharge from the walls of the ravine at elevations higher than the river stage, as shown in the photographs.

Both time series analysis and FO-DTS methods support the contention that the Silala River is indeed “a system of surface waters and groundwaters constituting by virtue of their physical relationship a unitary whole”, following the UNWC definition, and corresponds to an international watercourse. In addition, the insights obtained from these data were important in the development and calibration of a groundwater model of the system, which was used to confirm the conceptual understanding of the Silala River hydrogeology and to quantify the effects of historical channelization. Hence, thermal methods not only demonstrated that groundwater and surface waters interact throughout the Chilean reach of the Silala River, but also were valuable approaches to generate scientific evidence to support a major international legal dispute.

AUTHOR CONTRIBUTIONS

Francisco Suárez: Conceptualization (lead); formal analysis (equal); funding acquisition (lead); investigation (equal); methodology (equal); project administration (lead); supervision (lead); validation (equal); visualization (equal); writing – original draft (lead); writing – review and editing (lead). **Victoria Sandoval:** Data curation (equal); formal analysis

(lead); methodology (equal); writing – original draft (supporting); writing – review and editing (supporting). **Andrés Sarabia:** Data curation (equal); formal analysis (lead); methodology (equal); writing – original draft (supporting); writing – review and editing (supporting). **José F. Muñoz:** Funding acquisition (lead); investigation (equal); writing – original draft (supporting); writing – review and editing (supporting).

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CONFLICT OF INTEREST

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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